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Helberg

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(54) **SEPARATION OF BaCl₂ FROM CaCl₂
BRINE SOLUTION**

(71) Applicant: **THE CHEMOURS COMPANY FC,
LLC, Wilmington, DE (US)**

(72) Inventor: **Lisa E. Helberg, Middletown, DE (US)**

(73) Assignee: **THE CHEMOURS COMPANY FC,
LLC, Wilmington, DE (US)**

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Primary Examiner — Sally A Merkling

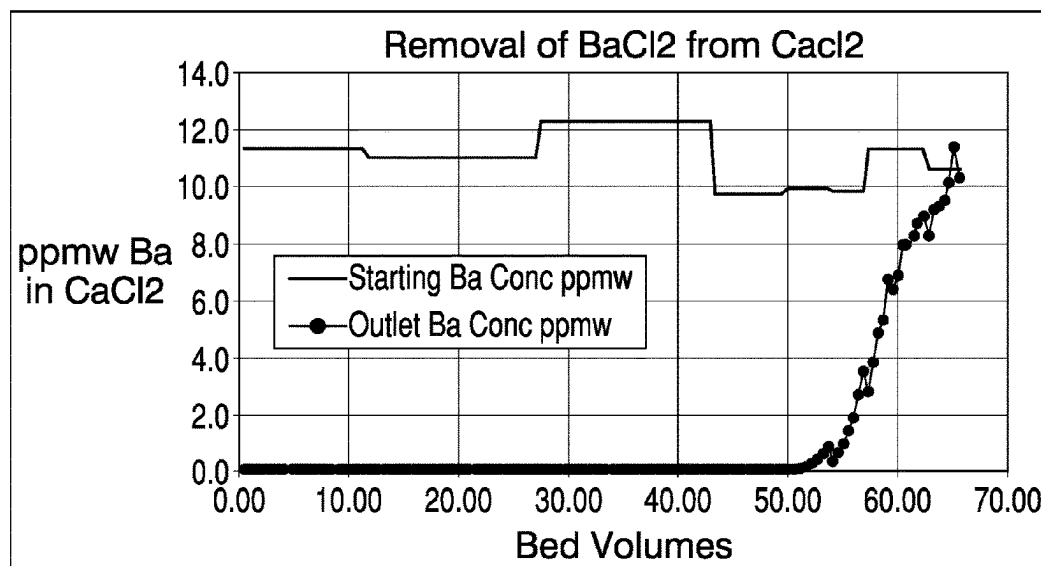
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ABSTRACT

The present invention relates to a process for separating
BaCl₂ from a CaCl₂ brine solution.

14 Claims, 2 Drawing Sheets



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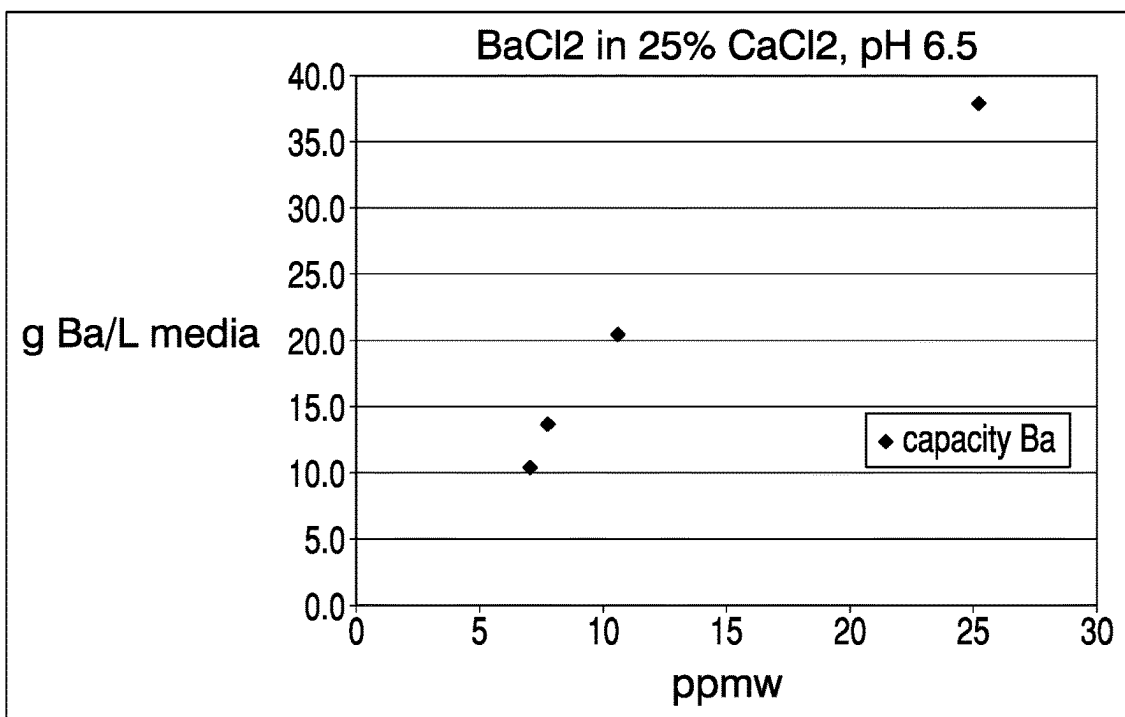
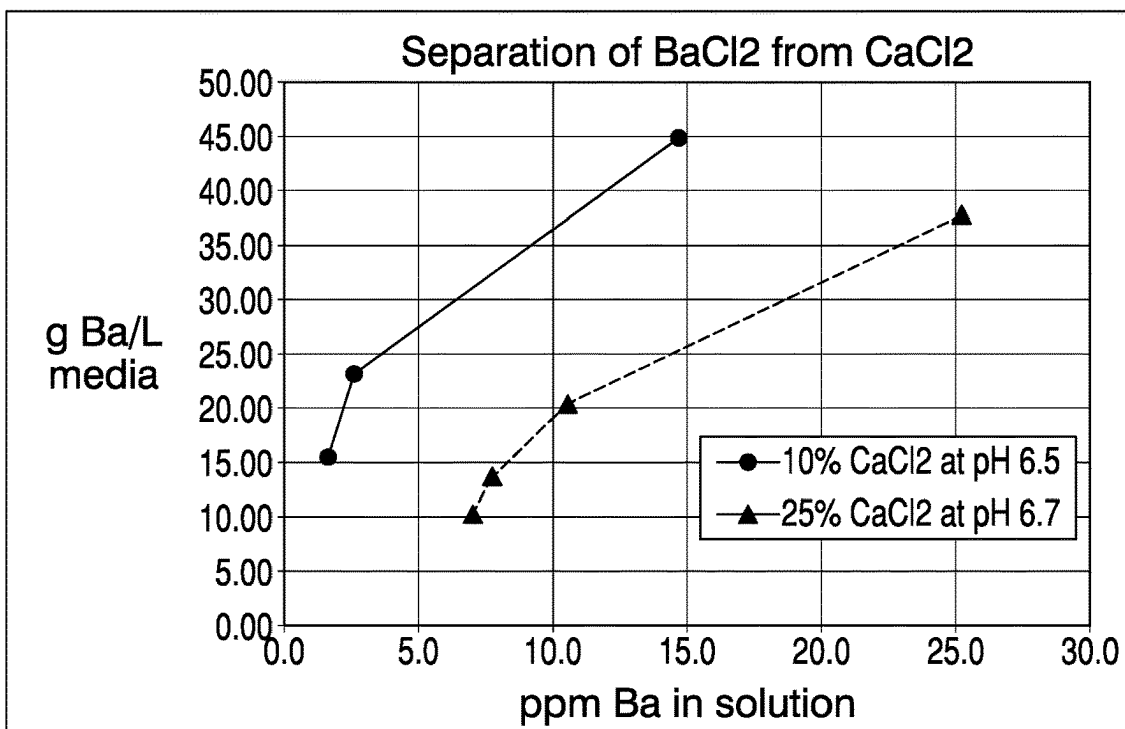
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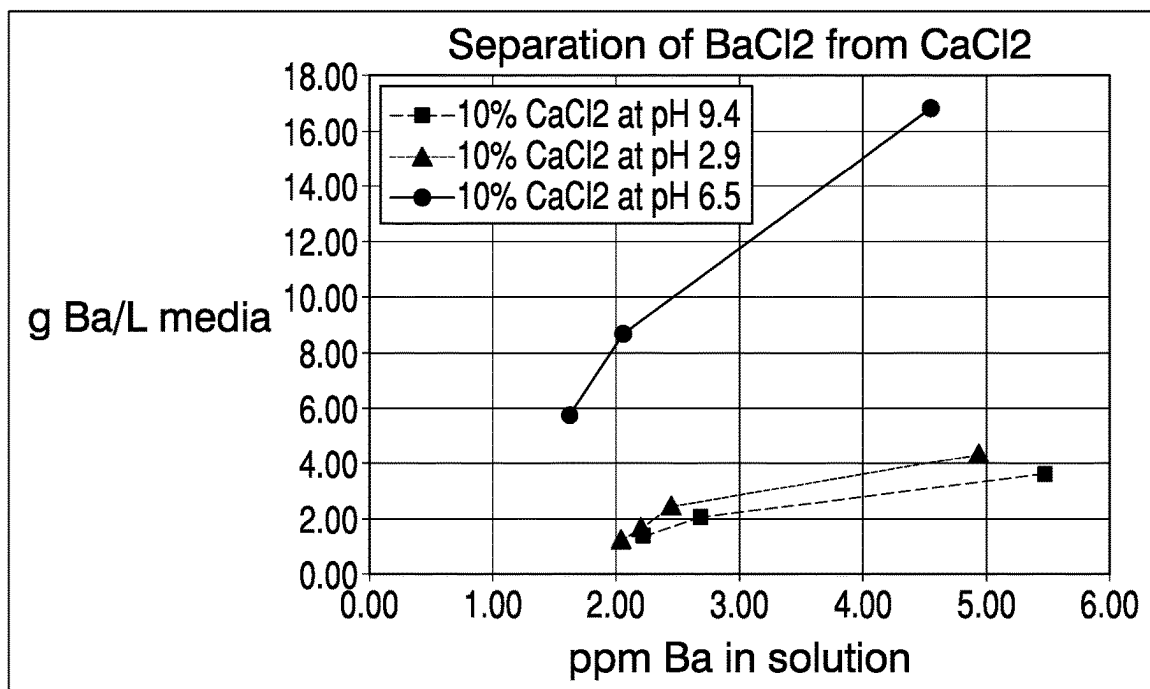
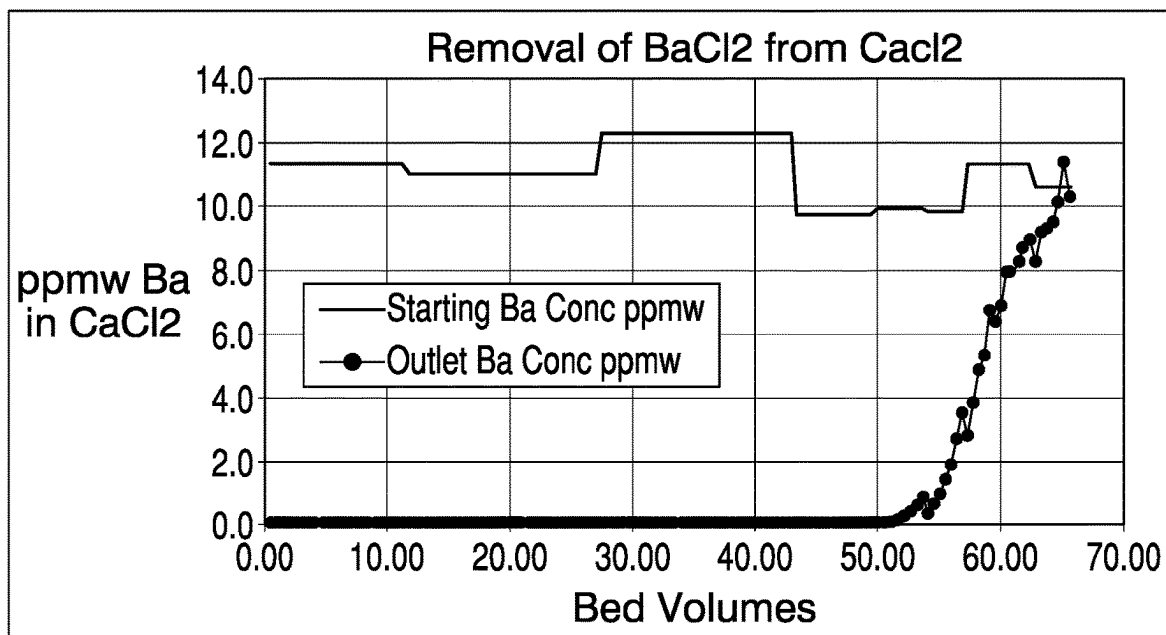
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**FIGURE 1****FIGURE 2**

**FIGURE 3****FIGURE 4**

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SEPARATION OF BaCl₂ FROM CaCl₂ BRINE SOLUTION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a national filing under 35 U.S.C. 371 of International Application No. PCT/US2020/042256 filed Jul. 16, 2020, and claims the benefit of priority of U.S. Provisional Application No. 62/875,120 filed Jul. 17, 2019, the disclosures of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

Background of the Invention

CaCl₂ is a widely used industrial chemical. It is produced primarily from three process routes: purification of natural CaCl₂ brines, neutralization of HCl with CaCO₃, or the Solvay Process which converts NaCl and CaCO₃ into Na₂CO₃ and CaCl₂. Each of these routes has advantages and disadvantages. CaCl₂ brine can be used directly in some applications or converted to a solid hydrated or anhydrous product. The CaCl₂ solutions are used in multiple industries including road de-icing, dust control, oil field production, cooling units or food applications. Each application has different quality requirements, but in the United States most require meeting Universal Treatment Standards (UTS) quality such as for road de-icing and oil field applications (see 40 CFR § 268.48—Universal Treatment Standards).

Acidic chloride streams are found as the by-product in many types of industrial scale processes. Neutralization of these streams with Ca(OH)₂, CaCO₃ or CaO has the opportunity to produce new sources of valuable CaCl₂ brine if the correct quality requirements can be met for the end use. One example is the iron chloride co-product stream produced from the Chloride Process in the manufacturing of TiO₂.

In the Chlorination step of the Chloride TiO₂ process, a ferrotitanium ore is chlorinated with a mixture of Cl₂, coke and heat to form TiCl₄ and a mixture of the other chlorinated metals that were present in the original ore. The composition of this mixture depends on the type of ore used such as ilmenite, slags or leucoxines, and the unique impurities in each ore body. These iron chlorides and other impurities are mixed with water to form a low pH aqueous solution. One of the options for this aqueous solution is to neutralize with a calcium base such as Ca(OH)₂ which leads to the formation of an iron hydroxide solid and a neutral or basic CaCl₂ stream. The majority of the impurities from the original iron chloride mixture are quantitatively removed from the resulting CaCl₂ brine solution.

From the UTS list of elements, one species of concern is Ba present as BaCl₂ in the CaCl₂ produced from this neutralization route. Ba is frequently an element of concern for quality in other neutralizations as well as in naturally occurring brine sources since the chemistry of Ba and Ca are similar.

Crystallization of the CaCl₂ would allow for separation of the BaCl₂ to meet the UTS standards, but this process is energy intensive and would reduce the savings and increase the capital requirements. For applications where a solid CaCl₂ product is not required, production of a suitable CaCl₂ solution through direct separation of the BaCl₂ in solution would allow for the potential of direct sales into UTS markets without crystallization. This separation directly

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from solution is very challenging due to the chemical similarities between Ba and Ca.

Crystallization and washing is a known solution for purifying CaCl₂. The separation of Ba and Ca ions from aqueous solution can be accomplished using ion exchange resin, but this technique cannot be applied to remove Ba ions from a Ca ion solution (CaCl₂ brine) because of the overwhelming Ca concentration and the ion exchange resin removes both types of +2 ions. CaCl₂ solution with BaCl₂ can be treated with CaSO₄ to form the less soluble BaSO₄ which can be removed from the CaCl₂ solution by filtration. This filtration is challenging due to the fine particle size of the BaSO₄ formed and its preference to form deposits inside process equipment. Additionally, this treatment does not allow the UTS goals to be met due to the solubility limits of BaSO₄ in CaCl₂. The presence of residual BaSO₄ in CaCl₂ product can also cause issues in use when the CaCl₂ solution is diluted because the solubility of BaSO₄ goes down and will precipitate, leaving residue build-up in process equipment.

BRIEF SUMMARY OF THE INVENTION

The invention results in a surprising adsorption of the BaCl₂ directly from CaCl₂ containing brine using a titanium containing material.

The invention comprises a method for purifying CaCl₂ brine containing BaCl₂. The method comprises the steps of contacting a CaCl₂ brine containing at least some BaCl₂ with a titanium containing material. Upon contacting the CaCl₂ brine with the titanium containing material, BaCl₂ is removed from the brine.

DESCRIPTION OF THE FIGURES

FIG. 1 is a graph showing the concentration of Metsorb® HMRG granules used in Example 1.

FIG. 2 is a graph showing the comparison of equilibrium removal of BaCl₂ from a 10% and 25% CaCl₂ according to Example 2.

FIG. 3 is a graph summarizing the conditions and results from the treatment of 10% CaCl₂ solution at pH 6.5 with different concentrations of Metsorb® HMRG granules according to Example 3.

FIG. 4 is a graph showing the results from Example 4.

DETAILED DESCRIPTION OF THE INVENTION

The invention results in a surprising adsorption of the BaCl₂ directly from CaCl₂ using a titanium containing material.

The invention comprises a method for purifying CaCl₂ brine containing BaCl₂. The method comprises the steps of contacting a CaCl₂ brine containing at least some BaCl₂ with a titanium containing material. Upon contacting the CaCl₂ brine with the titanium containing material the BaCl₂ is removed from the brine.

The titanium containing material can be contacted with the CaCl₂ brine in any suitable manner and under any conditions that will result in the removal of at least some BaCl₂ from the CaCl₂ brine. For example, the titanium containing material can be mixed into or with the CaCl₂ brine in a sufficient quantity to result in removal of at least some BaCl₂. Suitable weight ratios of the material are, for example, between 0.1 wt % and 5 wt % for batch separations. More preferable for batch separations is contacting the

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CaCl₂ brine with sequential dosing such as practiced in a resin in pulp configuration known to one skilled in the art. The contacting can also be done using a granular form of the titanium material in a column where the CaCl₂ brine is passed through a fixed bed either downflow or upflow with a suitable contact time controlled by the flow rate to allow the adsorption of the BaCl₂.

In an aspect of the invention the CaCl₂ brine solution has a concentration between about 10% to about 30% CaCl₂ at ambient temperature, based on the total weight of the brine solution.

In an aspect of the invention the CaCl₂ brine solution is contacted with the titanium containing material at a temperature between about 10° C. to about 75° C. In a further aspect of the invention the CaCl₂ brine solution is contacted with the titanium containing material at a temperature between about 20° C. to about 65° C.

The titanium containing material can be hydrated. Moreover, the titanium containing material can comprise TiO₂ or Ti(OH)₄, or a combination thereof. The titanium containing material can be provided in any suitable form, such as granular or powders, or a combination thereof.

Combining the titanium containing material with the brine is typically done at ambient pressure and temperature. The brine can be at any suitable pH. In an aspect of the invention the pH is in the range of from about 3 to about 9. In a further aspect of the invention the pH is in the range of from about 5.9 to about 6.9. In a still further aspect of the invention, the brine is at a pH of about 6.5.

The titanium containing material can be in any suitable form. When the titanium containing material is granules, the granules can have a particle size distribution suitable for loading into a column and passing an appropriate flow of the CaCl₂ of the desired concentration through the fixed bed with minimal backpressure. An example of a suitable size would be -16 mesh to 60 mesh material. Other suitable sizes that could be used in commercial equipment such as 8×30 or 12×40 are equally appropriate with the understanding of the relationship between the average surface area and capacity in service. In an aspect of the invention the titanium containing material can have a surface area above 180 m²/g.

In one aspect of the invention, the titanium containing material is a titanium containing material sold under the tradename Metsorb® HMRG Granular Media produced by Graver Technologies, Inc., located at 200 Lake Drive, Glasgow, DE 19702. Metsorb® HMRG is a hydrated titanium form sold commercially to remove heavy metals such as arsenic and lead from drinking water. Metsorb® HMRG is a crystalline titanium oxide (TiO₂) (anatase) with a moisture content of less than 7%, a particle size of from -16 mesh to +60 mesh, with a surface area of from 200-240 m²/g, a bulk density of 0.65 gram per cc (40 lbs./ft³), a pore volume of from 0.34 to 0.44 cm³/g, and an average pore size of from 64 to 84 Angstroms.

The titanium containing material can be contacted with the brine using any suitable method, such as mixing together, or by passing the brine through a column containing the titanium containing material. In an aspect of the invention, the column is of a suitable dimension and is packed with the Metsorb® HMRG granules. The CaCl₂ can be passed downflow through the packed bed with a suitable contacting time, under ambient pressure and temperature.

EXAMPLES

Example 1

A 25% CaCl₂ brine solution at pH 6.5 is spiked with 324 ppmw Ba as BaCl₂. The solution is divided into equal

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portions and contacted with differing amount of the Metsorb® HMRG granules. No wetting of the granules is required prior to contacting with the CaCl₂ solution. The solution is allowed to stir at room temperature for 24 hours, and the resulting concentrations of Ba in the product solutions are measured.

Table 1. Summary of Conditions and Results from the Treatment of 25% CaCl₂ solution at pH 6.5 with Different Concentrations of Metsorb® HMRG granules.

TABLE 1

Media weight (g)	Starting Measured Ba concentration (ppm)	End Measured Ba concentration (ppm)	Calculated capacity (g Ba/L media)
1.0344	324	25.2	37.84
2.0064	324	10.6	20.48
3.0287	324	7.77	13.74
4.0471	324	7.07	10.35

These concentrations are used to prepare the isotherm graph shown in FIG. 1. From the isotherm, an equilibrium capacity for the Ba can be calculated to be 47 g Ba/L Metsorb® HMRG granules for a starting 30 ppmw solution. This amount of capacity is well into the region that could make a commercially viable separation depending on the specifics of the process.

FIG. 1 shows an example of equilibrium removal of BaCl₂ from a 25% CaCl₂ solution at room temperature and pH 6.5 with Metsorb® HMRG granules.

Example 2

In a second example, following the same procedure as the results shown in Table 1 for Example 1, the starting CaCl₂ concentration is reduced from 25% to 10% CaCl₂. This example demonstrates that the concentration of the CaCl₂ solution has an impact on the capacity with higher CaCl₂ leading to lower capacity; however, the separation still occurs with significant removal. If the lower capacity is due to the higher viscosity leading to a lower mass transfer rate, higher contact time could increase capacity.

Table 2. Summary of Conditions and Results from the Treatment of 10% CaCl₂ solution at pH 6.5 with Different Concentrations of Metsorb® HMRG granules.

TABLE 2

Media weight (g)	Starting Measured Ba concentration (ppm)	End Measured Ba concentration (ppm)	Calculated capacity (g Ba/L media)
1.0213	360	14.7	44.87
2.0412	360	2.62	23.15
3.0195	360	1.67	15.50

FIG. 2 shows a comparison of equilibrium removal of BaCl₂ from a 10% and 25% CaCl₂ solution at room temperature and pH 6.5 with Metsorb® HMRG granules and similar starting concentrations.

Example 3

This example demonstrates that the pH of the CaCl₂ solution does have an impact on the equilibrium capacity. The Metsorb® HMRG granules have a natural pH near 6.5 since it is designed for operation in drinking water not a

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brine solution. The preferred embodiment is near pH 6.5. The separation of Ba is still feasible at a wide range of pHs with test results available between pH 3 and pH 9. The stability of the media would be of concern for long term operation outside that pH range.

Following the same procedure used in Example 1, a series of tests were done with BaCl₂ in 10% CaCl₂ solutions. In Table 3, the results are shown for a 10% CaCl₂ solution held at pH 6.5 and spiked with 136 ppmw Ba. Table 4, shows the same type of experiment at pH 2.9 and Table 5 shows the results at pH 9.4.

Table 3. Summary of Conditions and Results from the Treatment of 10% CaCl₂ solution at pH 6.5 with Different Concentrations of Metsorb® HMRG granules.

TABLE 3

Media weight (g)	Starting Measured Ba concentration (ppm)	End Measured Ba concentration (ppm)	Calculated capacity (g Ba/L media)
1.0217	136	4.55	16.80
2.0069	136	2.06	8.69
3.0704	136	1.63	5.75

Table 4. Summary of Conditions and Results from the Treatment of 10% CaCl₂ solution at pH 2.9 with Different Concentrations of Metsorb® HMRG granules.

TABLE 4

Media weight (g)	Starting Measured Ba concentration (ppm)	End Measured Ba concentration (ppm)	Calculated capacity (g Ba/L media)
1.0874	41	4.94	4.35
2.0652	41	2.45	2.47
3.0141	41	2.21	1.68
4.0573	41	2.04	1.24

Table 5. Summary of Conditions and Results from the Treatment of 10% CaCl₂ solution at pH 9.4 with Different Concentrations of Metsorb® HMRG granules.

TABLE 5

Media weight (g)	Starting Measured Ba concentration (ppm)	End Measured Ba concentration (ppm)	Calculated capacity (g Ba/L media)
1.0649	35	5.47	3.63
2.0219	35	2.69	2.08
3.0064	35	2.22	1.42

FIG. 3 shows a comparison of equilibrium removal of BaCl₂ from a 10% CaCl₂ solution at room temperature at a range of starting pHs with Metsorb® HMRG granules and similar starting concentrations.

Another common impurity found in CaCl₂ solution (and dry CaCl₂) is SrCl₂. The presence of SrCl₂ does not appear to impact the removal or capacity for BaCl₂, but the SrCl₂ is also not removed. The other expected impurity, RaCl₂ that could be present in CaCl₂ solutions might be expected to be removed in this process, but has not been determined at this time.

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Example 4

This Example demonstrates that the Metsorb® HMRG granules can be loaded into a column to allow for the treatment of a CaCl₂ solution continuously. In this example, the Metsorb® HMRG granules were loaded into a chromatography column and allowed to equilibrate at pH 6.5 using standard laboratory practices. A series of stock solutions of 10 wt % CaCl₂ were prepared using CaCl₂*2H₂O and DI water. The solution was spiked with 10 ppmw Ba, added as BaCl₂*2H₂O. Each solution was adjusted to pH 6.5 with NaOH before introduction to the column. CaCl₂ brine was pumped downflow through the column at a flow rate of 0.5 BV/hr to allow the large sized granules to remove the BaCl₂ effectively. The series of stock solutions were pumped through the bed until the inlet and outlet Ba concentrations were approximately equal. As shown in FIG. 4, for the first 50 Bed Volumes (BV), the Ba concentration was <100 ppbw in the outlet CaCl₂ solution. As more CaCl₂ brine was passed through the column, the measured outlet Ba concentration continued to rise until it reached an equilibrium value at 65 BV. This example shows that BaCl₂ can be removed to very low concentrations in CaCl₂ brine solutions allowing for purification to levels suitable for UTS applications or other types of application where low Ba levels are required without requiring a crystallization step and the high energy requirements needed.

What is claimed is:

1. A method for separating BaCl₂ from a CaCl₂ brine solution comprising contacting a titanium containing material with a BaCl₂ containing CaCl₂ brine solution, wherein the titanium containing material comprises TiO₂.
2. The method of claim 1, wherein the titanium containing material is hydrated.
3. The method of claim 1, wherein the titanium containing material further comprises Ti(OH)₄.
4. The method of claim 1, wherein the titanium containing material comprises granules.
5. The method of claim 1, wherein the titanium containing material comprises powder.
6. The method of claim 1, wherein the titanium containing material and the CaCl₂ brine solution are mixed together.
7. The method of claim 1, wherein the CaCl₂ brine solution is passed through a column packed with the titanium containing material in a granular form.
8. The method of claim 1, wherein the CaCl₂ brine solution is maintained at a pH from about 3 to about 9.
9. The method of claim 8, wherein the CaCl₂ brine solution is maintained at a pH from about 5.9 to about 6.9.
10. The method of claim 9, wherein the CaCl₂ brine solution is maintained at a pH of about 6.5.
11. The method of claim 1, wherein the CaCl₂ brine solution has a concentration between about 10 wt. % to about 30 wt. % CaCl₂ at ambient temperature, based on the total weight of the brine solution.
12. The method of claim 1, wherein the CaCl₂ brine solution is contacted with the titanium containing material at a temperature between 10° C. and 75° C.
13. The method of claim 12, wherein the CaCl₂ brine solution is contacted with the titanium containing material at a temperature between 20° C. and 65° C.
14. The method of claim 1, wherein the titanium containing material has a surface area above 180 m²/g.

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